

# Vincent Zaballa | Ph.D.

Irvine, CA

 [vz415.github.io](https://github.com/vz415) •  [vz415](https://orcid.org/vz415)

## Education

### University of California, Irvine

*Ph.D., Biomedical Engineering*

Thesis: "Machine Learning Methods for Posterior Approximation, Mutual Information Optimization, and Integrating Structural and Systems Biology in Drug Discovery."

Irvine, CA

2019–2026

### Imperial College London

*M.Res., Bioengineering*

London, UK

2015–2016

### Texas A&M University

*B.S. & M.Eng., Biomedical Engineering, Minor in Electrical Engineering, cum laude*

College Station, TX

2010–2015

## Research Interests

My research develops score-based diffusion models and normalizing flows for scientific applications. I am particularly interested in (1) diffusion transformer architectures for biological systems, including principled fine-tuning, compositional sampling, and hierarchical inference over structured biological data; (2) connecting generative modeling with Bayesian experimental design to enable simulation-efficient active learning under epistemic uncertainty; and (3) integrating models across biological scales to enable multi-modal generative approaches for perturbation modeling, molecular design, and structured biological inference. I enjoy building theoretical foundations and translating them into working code, fast, and contributing to open-source tools when possible (Radical Numerics, LFIAX, JAX Optax).

## Experience

### University of California, Irvine

*Graduate Research Assistant*

Irvine, CA

Sept. 2019–2026

- **[BOED-agents]** Building an agentic system for simulator-agnostic Bayesian optimal experimental design that adaptively selects models, infers parameters, and proposes informative experiments under simulator uncertainty and fixed experimental budgets.
- **[BMP Simformer]** Developed novel score-based diffusion transformer foundation model to expand the predictive capabilities of systems biology. (*NeurIPS workshop*/In submission at ICLR 2027)
- **[BMP Simformer]** Implemented a principled fine-tuning method for score-based diffusion models using full diffusion rollouts regularized by the reverse KL divergence to the pretrained model, leveraging the Girsanov theorem to regularize over path measures rather than naïve parameter-space distances — reducing posterior inference RMSE by 97% ( $14.43 \rightarrow 0.389$ ) and introducing a novel model misspecification technique  $\mathcal{A}(\theta, \xi)$  that indicates where an expert simulator may be misspecified.
- **[BMP Simformer]** Derived compositional posterior sampling diffusion coefficient  $g_n(t)$  that improves accuracy & numerically stabilizes Langevin sampling of continuous-time diffusion models to  $n$ -dependent observations, and developed Hierarchical Blockwise Diffusion Sampling (HBDS) for joint inference of shared parameters and group-specific nuisance variables — compressing posterior inference from 10,400 to 71 random variables. (Extends *Geffner et al., ICML 2023*.)
- **[SBI-BOED]** Proved that maximizing an InfoNCE mutual information bound is equivalent to jointly minimizing KL divergence of a surrogate likelihood and optimizing experimental designs, enabling Bayesian optimal experimental design for *non-differentiable* simulators with 52% better predictive accuracy and  $\sim 50\times$  fewer simulator calls than competing methods. (*coded in LFIAX*/Accepted at UAI 2026)
- **[SBI-BOED]** Developed a simulation-efficient active learning scheme in SBI, using MC-dropout uncertainty estimates to prioritize simulator queries in InfoNCE, reducing posterior calibration error up to 50% over random sampling under fixed simulation budgets.
- Developed new method to perform diffusion guidance applied to small molecule generation in systems biology and structure-based drug design (*coded in PyTorch*). (*NeurIPS MLSB, 2024*).
- Demonstrated a new approach to integrating structural and systems biology information, improving predictions by 1.34% without collecting additional experimental data (*ICML ML4LMS, 2024*).
- Modeled the intractable likelihood of a high-dimensional dataset of the Bone Morphogenetic Protein (BMP) signaling pathway using normalizing flows (*NeurIPS GenBio, 2023*).
- Created a novel approach for optimizing Mutual Information & experimental designs in Simulation-based Inference, improving feasibility in high-dimensional design optimization (*ICML DAE, 2023*).
- Developed automated biological model selection algorithm with 166–298% better accuracy over baseline methods (*ICML CompBio, 2022*).
- Improved average experimental design efficiency by 225% using surrogate probabilistic modeling of physics-based biological models and Bayesian optimization (*NeurIPS LMRL, 2021*).

## Fannin Innovation Studio

Project Manager

Houston, TX

June 2018–Aug. 2019

- Managed commercialization and R&D of a pioneering fetal surgery device, resulting in a \$2 million SBIR Phase II Grant.
- Managed two hardware engineers and one software engineer – introduced new quality control computer vision software in OpenCV.

## Drip

Data Scientist

Colorado Springs, CO

April 2017–Oct. 2017

- Developed analytics and machine learning algorithms to draw insights from Colorado water data records within a SaaS application.

## Cambiando Paradigmas

Data Analyst

La Paz, Bolivia

Jan. 2017–June 2018

- Improved returns from renewable energy investments by 4.5x over previous portfolio, using random forests and optimization.

## Imperial College London

Whitaker Fellow

London, UK

Oct. 2015–Oct. 2016

- Designed a novel orthopedic implant that reduces typical bone strain by 90% (validated via finite element analysis).

## Biomedical Device Lab, Texas A&M

Research Assistant

College Station, TX

Jan. 2013–May 2015

- **Undergraduate Research Scholar Thesis:** “Characterization Of Automatically Reticulated Shape Memory Polymer Foams.”
- Created an automated system for processing Shape Memory Polymer (SMP) foams, reducing manufacturing time by 4.5 hours.

## Publications

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**2026:** V. Zaballa, E. Bourgain-Chang, “MetaBOED: Agentic Control of Inference and Experimental Design under Simulator Uncertainty.” *In preparation.*

**2026:** V. Zaballa, E. Hui, “Scalable Diffusion SBI for Compositional Inference under Simulator Misspecification.”

**2026:** V. Zaballa, E. Hui, “Optimizing Likelihoods via Mutual Information: Bridging Simulation-Based Inference and Bayesian Optimal Experimental Design.” [UAI 2026.](#)

**2024:** V. Zaballa, E. Hui, “Systems-Structure-Based Drug Design.” *NeurIPS MLSB Workshop, 2024.* [Preprint.](#)

**2024:** V. Zaballa, E. Hui, “Reducing Uncertainty Through Mutual Information in Structural and Systems Biology.” *ICML ML4LMS Workshop, 2024.* [Preprint.](#)

**2023:** V. Zaballa, E. Hui, “Approximation of Intractable Likelihood Functions in Systems Biology via Normalizing Flows.” *NeurIPS GenBio Workshop, 2023.* [Preprint.](#)

**2023:** V. Zaballa, E. Hui, “Stochastic Gradient Bayesian Optimal Experimental Designs for Simulation-based Inference.” *ICML Differentiable Almost Everything Workshop, 2023.*

**2022:** V. Zaballa, E. Hui, “An Optimal Likelihood Free Method for Biological Model Selection.” *ICML Workshop on Computational Biology, 2022.* [Preprint.](#)

**2021:** V. Zaballa, E. Hui, “Optimal Design of Experiments for Simulation-Based Inference of Mechanistic Acyclic Biological Networks.” *NeurIPS Learning Meaningful Representations of Life Workshop, 2021.* [Preprint.](#)

**2015:** V. Zaballa, D. Friedrichs, “A Novel Tissue Welding Device using Low-Voltage Far-Field Coaxial Electrospinning.” *ASME Design of Medical Devices Conference Journal, Jan. 2015.*

## Invited Talks & Presentations

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**Dec. 2025:** “Diffusion transformers as novel foundation models for systems biology.” *Neural Information Processing Systems (NeurIPS) Latinx in AI Workshop, San Diego, CA.*

**Apr. 2025:** “Optimizing Likelihoods via Mutual Information: Bridging Simulation-Based Inference and Bayesian Optimal Experimental Design.” *Lightning Talk at Stanford Experimental Design: AI for Science 2025, Palo Alto, CA.*

**Apr. 2022:** “Optimal Design of Experiments for Simulation-Based Inference of Mechanistic Acyclic Biological Networks.” *Lightning Talk at UCLA SoCalSysBio 2022, Los Angeles, CA.*

## Software

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Contributor, [Radical Numerics](#) Discrete Diffusion Language Modeling (2025) - Implemented Entropy-Bounded sampler

Contributor, [JAX Optax](#) optimization library (2023–2024)

Creator, [LFIAX](#): JAX conditional normalizing flows (2022)

## Teaching Experience

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### Engineering Analysis/Design: Data Analysis

Teaching Assistant

University of California, Irvine

Winter 2026

- Led recitation sections and addressed questions in a sophomore-level data analysis and programming course.

### Biology 50A: Cell and Molecular Engineering

Teaching Assistant

University of California, Irvine

Winter 2025

- Led recitation sessions and addressed questions in a sophomore-level biology and synthetic biology class.

### Maths 2

Teaching Assistant

Imperial College London

Oct.–Dec. 2015

- Led recitation sessions in multivariate calculus and differential equations for second-year students.

## Academic Service

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**2021–2025:** Reviewer, AISTATS (2022, 2023, 2025, 2026)

**2025–2026:** Reviewer, NeurIPS (2025, 2026)

**2024–2026:** Reviewer, ICLR (2025, 2026, 2027)

**2022:** Contributor, *Probabilistic Machine Learning: Advanced Topics* by Kevin Murphy

## Funding

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**2025:** NSF Innovation Corps (I-Corps) grant: \$50,000

**2022:** NIH F31 Ruth R. L. Kirschstein Research Service Award Fellowship: \$125,000

## Awards & Fellowships

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**2022:** ARCS Scholar Award (\$10,000)

**2021:** Google Computer Science Research Mentorship Program (Google mentor)

**2021:** Summer Inclusive Excellence Award – *for promoting minority access to undergraduate research* (\$5,000)

**2017:** Go Code Colorado Hackathon Winning Team (\$25,000)

**2015–2016:** Whitaker International Fellowship (\$75,000)

**2015:** Imperial College London Postgraduate Award for Outstanding Students (\$10,000)